

Shear force and Bending Moment.

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Beam: It is a structural member whose longitudinal dimension is large compared to transverse dimension. The beam is supported along its length and is acted upon by a system of loads at right angles to its axis. Due to the external loads and couples, shear force and bending moment develop at any section of the beams. Therefore for design of beams information about shear force and bending moment is desired.

Types of beams: The following are the important types of beams

1 Cantilever beam;

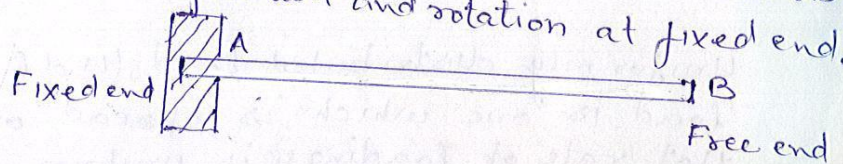
3 Overhanging beam

5 Continuous beam.

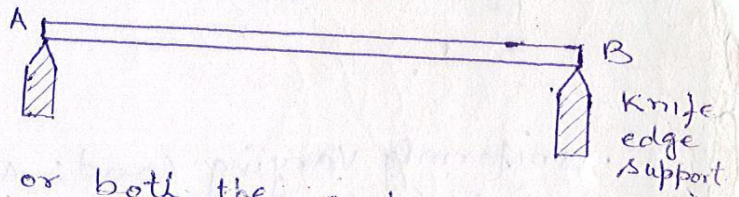
2 Simply supported beam

4 Fixed beam

Cantilever beam: A beam which is fixed at one end and free at the other end is called a cantilever beam as shown in fig. There is no deflection and rotation at fixed end.



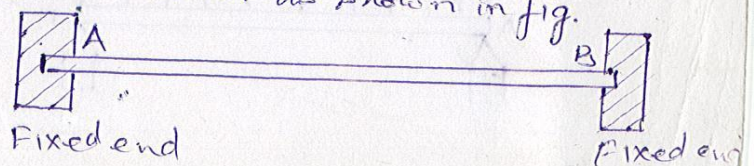
Simply supported beam: A beam which is supported or resting freely on the supports at its both ends is called simply supported beam. There is no deflection or displacement of the beam at the ends.



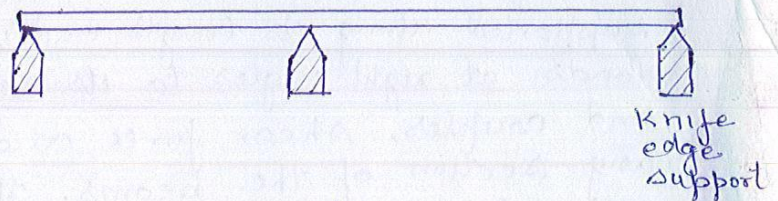
Overhanging beam: If one or both the ends of a beam are projected beyond the supports such a beam is called overhanging beam as shown in fig.



Fixed beam: A beam whose both ends are fixed or embedded in the wall or concrete is called a fixed beam as shown in fig.



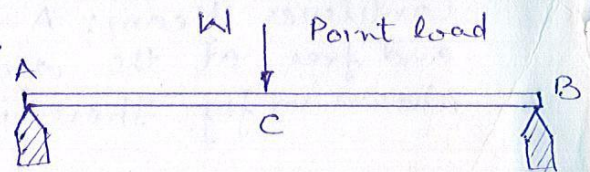
Continuous beam: A beam which is provided more than two support as shown in fig is called a continuous beam.



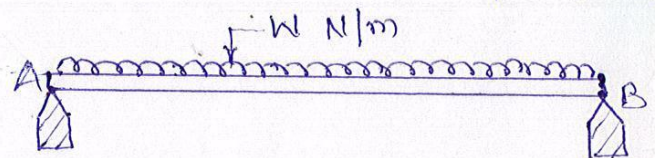
Types of Loads: A beam is normally horizontal and the loads acting on the beam are generally vertical. The following are the important types of loads acting on a beam.

- 1 Concentrated load or Point load
- 2 Uniformly distributed load
- 3 Uniformly varying load.

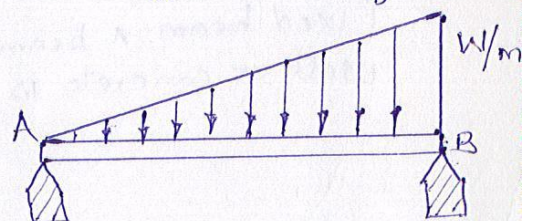
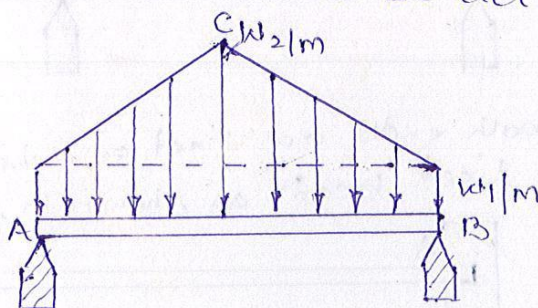
Concentrated load: A concentrated load is one which is considered to act at a point, as shown in fig.



Uniformly distributed load (U.d.l): A uniformly distributed load is one which is spread over a beam in such a manner that rate of loading w is uniform along the length i.e. each unit length is loaded to the same rate. The rate of loading is expressed in N/m run. For solving the numerical problems the total uniformly distributed load is converted into point load acting at centre of u.d.l.

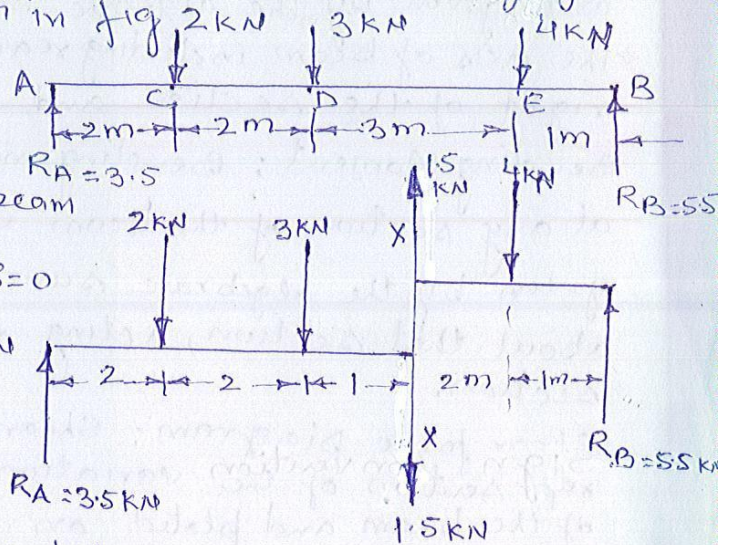


Uniformly Varying load: A uniformly varying load is one which is spread over a beam in such a way that the rate of loading varies from point to point along the beam as shown in fig. For solving numerical problems the total load is equal to the area of the load diagram and the total load is assumed to be act at the C.G. of the diagram.



Concept of Shear force and Bending Moment. 3/3

Consider a beam AB simply supported and carrying several pt loads as shown in fig



Now find out reactions

Considering the equilibrium of beam

$$\sum M_A = 2 \times 2 + 3 \times 4 + 4 \times 7 - R_B \times 8 = 0$$

$$8R_B = 44 \quad \text{or} \quad R_B = 5.5 \text{ kN}$$

$$\sum M_B = 4 \times 1 + 3 \times 4 + 2 \times 6 = 8R_A$$

$$R_A = \frac{28}{8} = 3.5 \text{ kN}$$

imaginary

Now take a section x-x at any distance from pt A (Let at 5m from A) which divide the beam into two parts one to the left & other to the right of the section

Now considering the equilibrium of forces on each portion of the beam the net resultant vertical forces:

To the Left of the Section: $F_{\text{left}} = 2 + 3 - 3.5 = 1.5 \text{ kN}$ acting down ward

To the right of Section: $F_{\text{right}} = 4 - 5.5 = -1.5 \text{ kN}$ acting upward

The forces to the right side and to the left side of the sect. are equal in magnitude and opposite in direction. It is concluded from here that section x-x is subjected to a force

1.5 kN which is trying to shear the beam at this section. Therefore this force is called shear force at section x-x (either right or left to sect. x-x)

Considering Equilibrium of moments on each side of the section

$$M_{\text{left}} = 3.5 \times 5 - 2 \times 3 - 3 \times 1 = 8.5 \text{ kNm} \quad \text{clockwise}$$

$$M_{\text{right}} = 5.5 \times 3 - 4 \times 2 = 8.5 \text{ kNm} \quad \text{anticlockwise}$$

The moments to the right and left side of the section are equal in magnitude and opposite in direction

It is concluded that section x-x is acted upon by a moment of 8.5 kNm. This moment is trying to bend the beam that is why it is called bending moment.

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Shear force: Shear force is defined as the force, at any section of the beam, which is trying to shear off the beam at that section and given by the algebraic sum of all the forces acting normal to the axis of beam including reactions either to the left or to the right of the section.

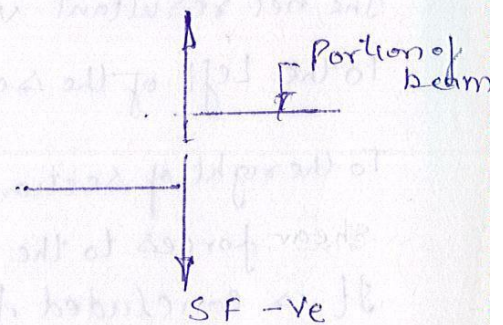
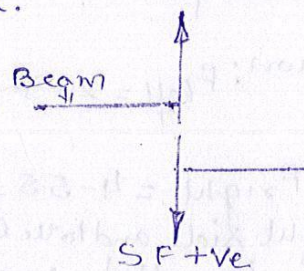
Bending Moment; Bending moment is defined as the moment at any section of the beam which tends to bend the beam and given by the algebraic sum of the moment of all the forces - including reactions about that section, acting either to the left or to the right of the section.

Shear force Diagram; Shear force diagram is the graphical representation of the variation of shear forces along the length of the beam and plotted on the base line by plotting +ve above and -ve value below the baseline. and abbreviated as S.F.D.

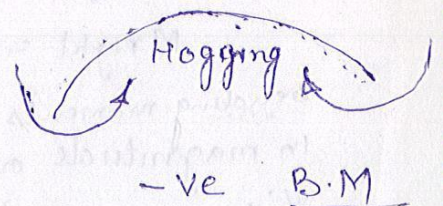
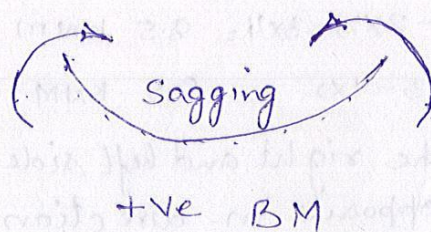
Bending moment Diagram (BMD); BMD is the graphical representation of the variation of bending moment along the length of the beam and plotted same as SFD, w.r.t base line.

Sign convention;

Shear force is taken +ve if it tends to move ^{either} the right portion downward or the left portion upward w.r.t. reverse portions. and vice versa.



Bending is taken +ve if it tends to sag (concave upward) the beam and is taken -ve if it tends to hog the beam. As shown in fig

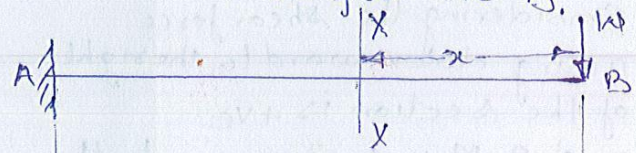


Important points for drawing SFD & BMD.

1. Consider the left or the right portion of the section.
2. Take the algebraic ~~sum~~ sum of forces including reactions normal to the beam on one portion.
If right portion of the section is chosen, a force on the right portion acting downwards is +ve while acting upwards is -ve.
If left portion of the section is chosen a force on the left portion acting downwards is -ve while a force acting upwards is +ve.
3. The +ve values of SFD & B.M are plotted above the base line and -ve values below the base line.
4. The SFD ^{will} increase or decrease suddenly i.e. by a vertical st line at a section where there is a vertical point load.
5. The S.F between any two vertical loads will be constant and hence the S.F D. between two vertical loads will be horizontal.
6. The bending moment at two supports of a simply supported beam and at free end of the cantilever beam will be zero.
7. At the fixed end of the cantilever SFD & B.M will be maximum.
8. Under UDL the SFD is inclined st line and B.M.D will be a curve.
9. For UVL, SFD is parabolic of second degree.
10. The B.M.D for UDL (triangular or trapezium) will be parabola of third degree.
11. The portion in which S.F is constant B.M curve is st. line.
12. Point of contra flexure is a pt where B.M. changes sign.

S.F.D and B.M.D for a cantilever beam with a pt. load at the free end.

A cantilever beam AB of length L , fixed at A and free at end B having a pt load w at the free end B.



Take a Sect. $x-x$ at a distance x from the free end B. S.F. at $x-x$ = algebraic sum of forces to the right of this section

$F_x = +w$ (considering force to the right to the section +ve)
S.F. will be constant between A & B as there is no other force between

A & B. Hence S.F. will be constant through out the length AB

S.F.D may be drawn by plotting the values of S.F. at diff. pts above the base line as shown in fig. The B.M. at the section $x-x$

is given by (considering clock wise moment rt to the Sect. -ve)

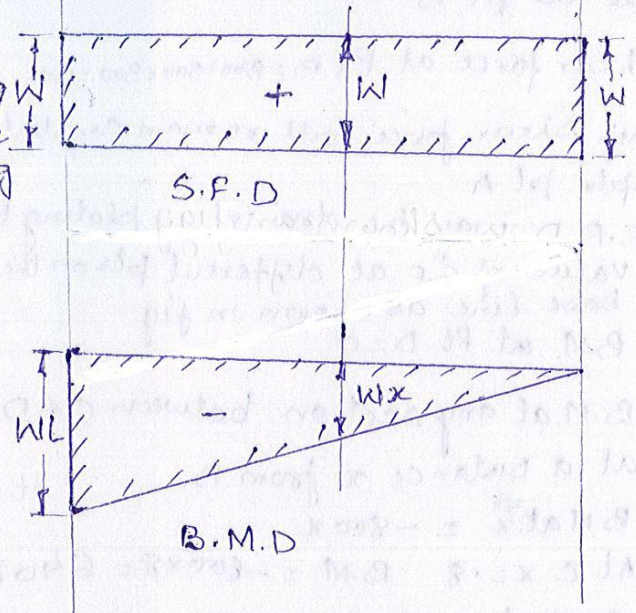
$$\therefore M_x = -wx$$

At B $x=0$ Hence B.M. = 0

At A $x=L$ " B.M. = $-wL$

B.M. follows the 1st line law

B.M.D may be drawn by plotting the values of B.M. at different points below the base line as shown in fig.



A cantilever beam of length 2m carries the point loads as shown in fig. Draw S.F.D & B.M.D for the beam.

Considering the force acting down ward to the right of the section is +ve and B.M. anti clockwise to the right of the section is +ve

SF at Pt D is 800N & remains constant just to the right of point C

Shear force at Pt C = $800 + 500 = 1300$ and remains constant upto just right the the pt B

Shear force at Pt B = $800 + 500 + 300 = 1600$

this shear force will remain constant upto pt A

S.F.D may be drawn by plotting the value of S.F. at different pts on the base line as shown in fig
B.M. at Pt D = 0

B.M. at any section between C & D

at a distance x from D.

At D $x=0$ $BM=0$

B.M. at $x = -800x$

At C $x=1.8$ $B.M. = -800 \times 1.8 = 1440 \text{ KNm}$

B.M. at a section between B and C at a distance x from D will be

$M_x = -800x - 500(x-0.8)$

Bending moment between B & C varies by a st line law

B.M. at B when $x=1.5\text{m}$

$$= -800 \times 1.5 - 500 \times 0.7 = 1550 \text{ KNm}$$

B.M. at a section between A & B at a distance x from D

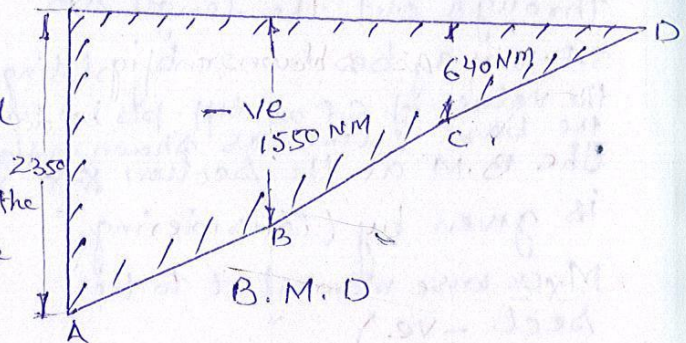
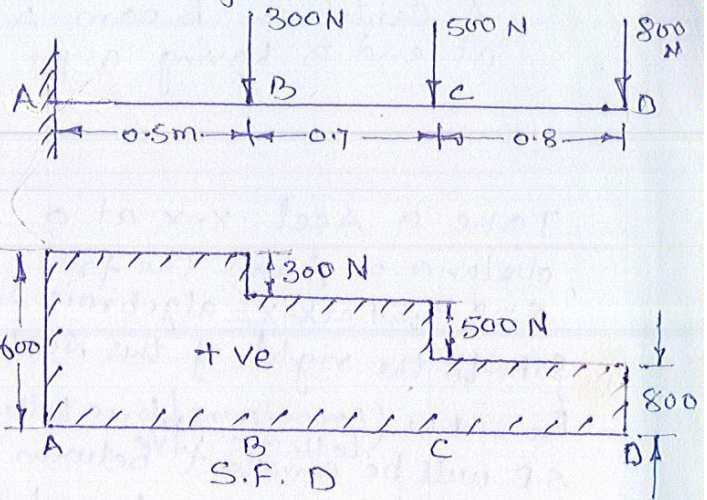
$$= -800x - 500(x-0.8) - 300(x-1.5)$$

Bending moment here also varies by st line law.

At A $x=2\text{m}$

$$B.M. \text{ at A } = -800 \times 2 - 500 \times 1.2 - 300 \times 0.5 = 2350 \text{ KNm}$$

B.M.D may be drawn by plotting the values of B.M. at different points below the base line and joining them as shown in fig



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S.F.D and B.M.D for a cantilever beam carrying a u.d.l of $w/\text{unit length}$ over the entire length of the cantilever

Take a section X-X at a distance x from free end considering the right portion of the section for which $\downarrow +ve$.

S.F at the section X-X = $+wx$

At B $x=0$ $SF = 0$

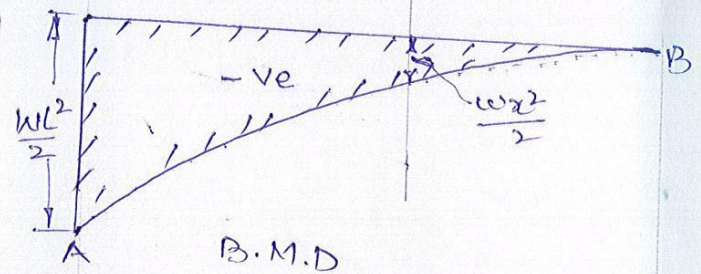
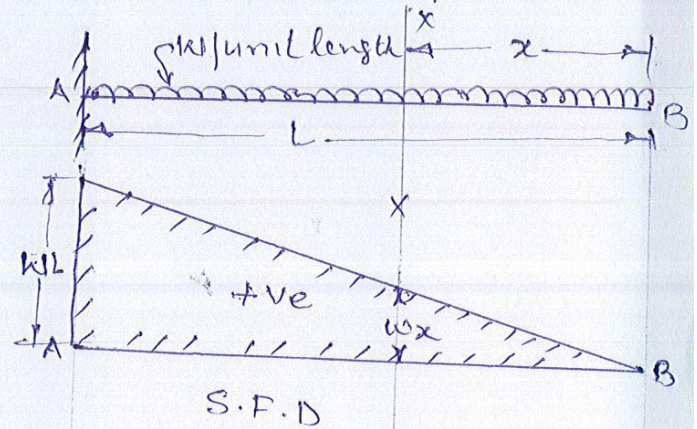
At A $x=L$ $SF = wL$

Bending Moment at section X-X

$$M_x = -wx \times \frac{x}{2} = -\frac{wx^2}{2}$$

At B $x=0$ $BM = 0$

At A $x=L$ $BM = -\frac{wL^2}{2}$



S.F.D & B.M.D for a cantilever beam carrying gradually varying load.

Taking a section X-X from B at a distance x & considering $\downarrow$ forces + to the right of the sect.

$$SF \text{ at section X-X} = \frac{wx}{L} \times \frac{x}{2} = \frac{wx^2}{2L}$$

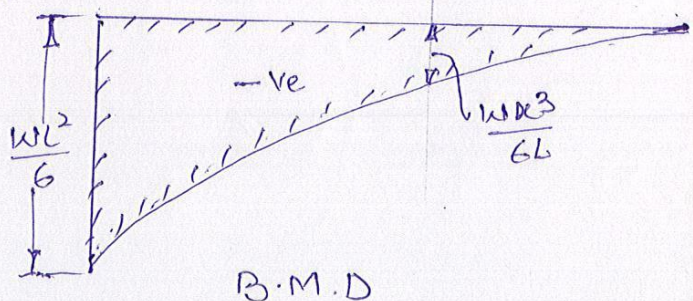
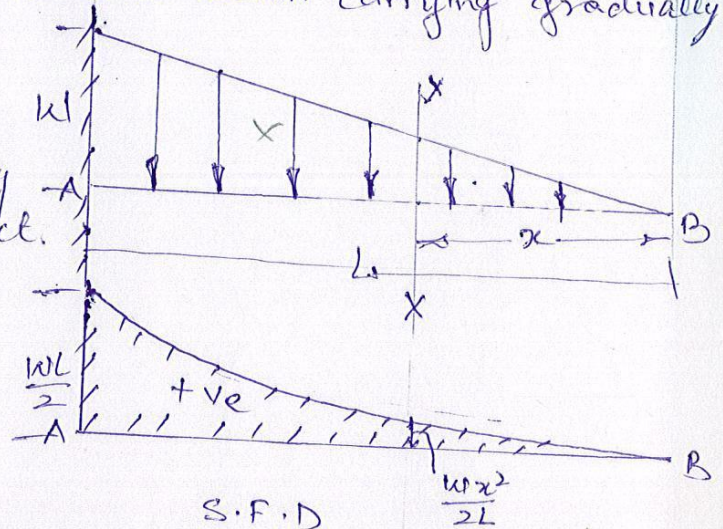
At B $x=0$ $SF = 0$

At A $x=L$ $SF = \frac{wL^2}{2L} = \frac{wL}{2}$

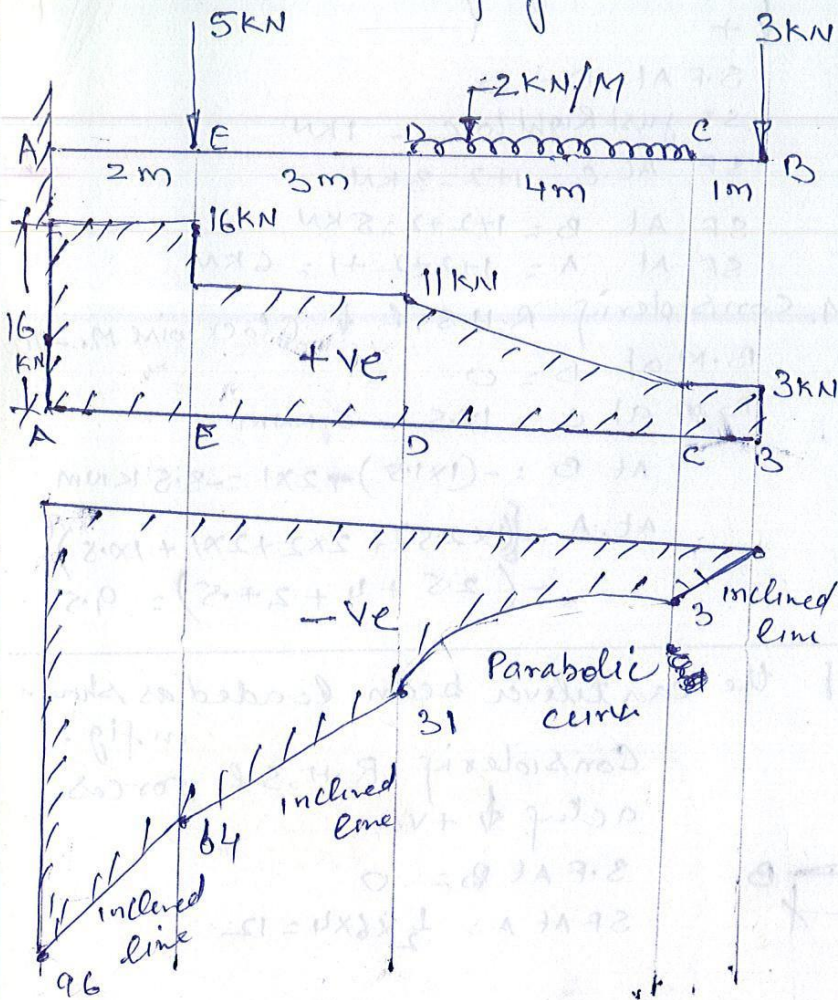
$$B.M. \text{ at X-X} = \frac{wx^2}{2L} \times \frac{x}{3} = \frac{wx^3}{6L}$$

At B $x=0$ $B.M = 0$

At A $x=L$ $B.M = \frac{wL^3}{6L} = \frac{wL^2}{6}$



Draw the S.F.D & B.M.D of a cantilever beam load as shown in fig.



Considering R.H.S and taking force $\downarrow$ +ve & $\uparrow$ -ve

$$S.F. \text{ At } B = +3 \text{ kN}$$

$$S.F. \text{ At } C = +3 \text{ kN}$$

$$" \text{ " } D = +3 + (2 \times 4) = 11 \text{ kN}$$

$$" \text{ approaches to } E = 11 \text{ kN}$$

$$" \text{ " } A \text{ to } E = 11 + 5 = 16 \text{ kN}$$

$$" \text{ " } A = 16 \text{ kN}$$

B.M. Clockwise -ve

$$B.M. \text{ at } B = 0$$

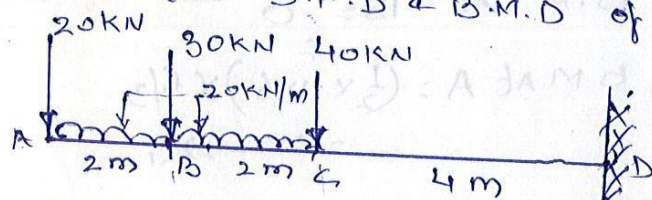
$$\text{At } C = -(3 \times 1) = -3 \text{ kNm}$$

$$D = -(3 \times 5) + (2 \times 4) \times 2 = -31 \text{ kNm}$$

$$E = -\left[(3 \times 8) + (2 \times 4) \times 5 \right] = -64 \text{ kNm}$$

$$A = -\left[(3 \times 10) + (2 \times 4) \times 7 + (5 \times 2) \right] = -96 \text{ kNm}$$

Draw the S.F.D & B.M.D of the cantilever beam loaded as shown in fig. Considering L.H.S & forces $\downarrow$ -ve



$$S.F. \text{ At } A = -20 \text{ kN}$$

$$" \text{ just left of } B = -(20 + 40) = -60 \text{ kN}$$

$$S.F. \text{ At } B = -(20 + 40 + 30) = -90 \text{ kN}$$

$$S.F. \text{ just left of } C = -(20 + 30 + 80) = -130 \text{ kN}$$

$$S.F. \text{ At } D = -(20 + 30 + 80 + 40) = -170 \text{ kN}$$

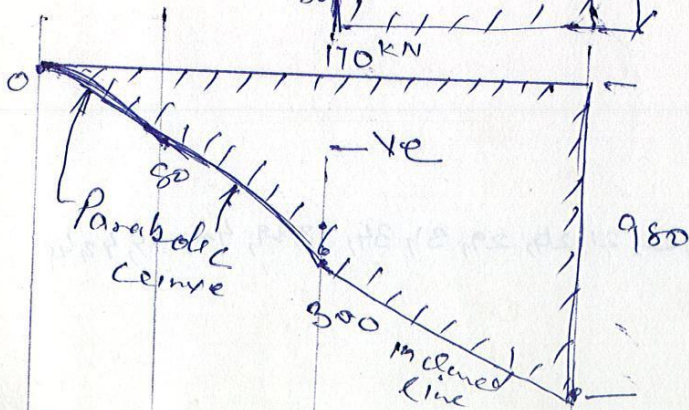
$$" \text{ At } D = -170 \text{ kN}$$

$$B.M. \text{ At } A = 0$$

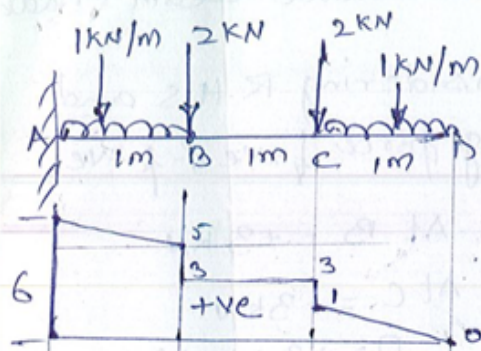
$$\text{At } B = -(20 \times 2 + (20 \times 2) \times 1) = -80$$

$$\text{At } C = -(20 \times 4 + 30 \times 2 + ((20 \times 4) \times 2)) = -(80 + 60 + 160) = -300 \text{ kNm}$$

$$\text{At } D = -(20 \times 8 + (30 \times 6) + 40 \times 4 + (20 \times 4) \times 6) = -(160 + 180 + 160 + 480) = -980 \text{ kNm}$$



Draw the S.F.D & B.M.D of a cantilever beam loaded as shown



Considering R.H.S & forces acting

↓ +

$$\text{S.F. at D} = 0$$

$$\text{SF just Right to C} = 1 \text{ kN}$$

$$\text{SF at C} = 1 + 2 = 3 \text{ kN}$$

$$\text{SF at B} = 1 + 2 + 2 = 5 \text{ kN}$$

$$\text{SF at A} = 1 + 2 + 2 + 1 = 6 \text{ kN}$$

B.M. Considering R.H.S. & ↓ Clockwise M. -ve

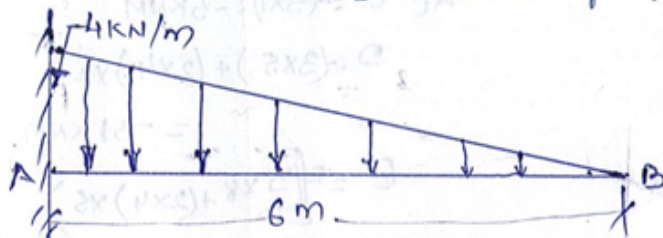
$$\text{B.M. at D} = 0$$

$$\text{B.M. at C} = 1 \times 1 = 1 \text{ kNm}$$

$$\text{At B} = -(1 \times 1.5) + 2 \times 1 = -3.5 \text{ kNm}$$

$$\begin{aligned} \text{At A} &= -\{(1 \times 2.5) + 2 \times 2 + 2 \times 1 + 1 \times 1\} \\ &= -(2.5 + 4 + 2 + 1) = -9.5 \end{aligned}$$

Draw the S.F.D & B.M.D of the cantilever beam loaded as shown



Considering R.H.S & forces acting ↓ +ve

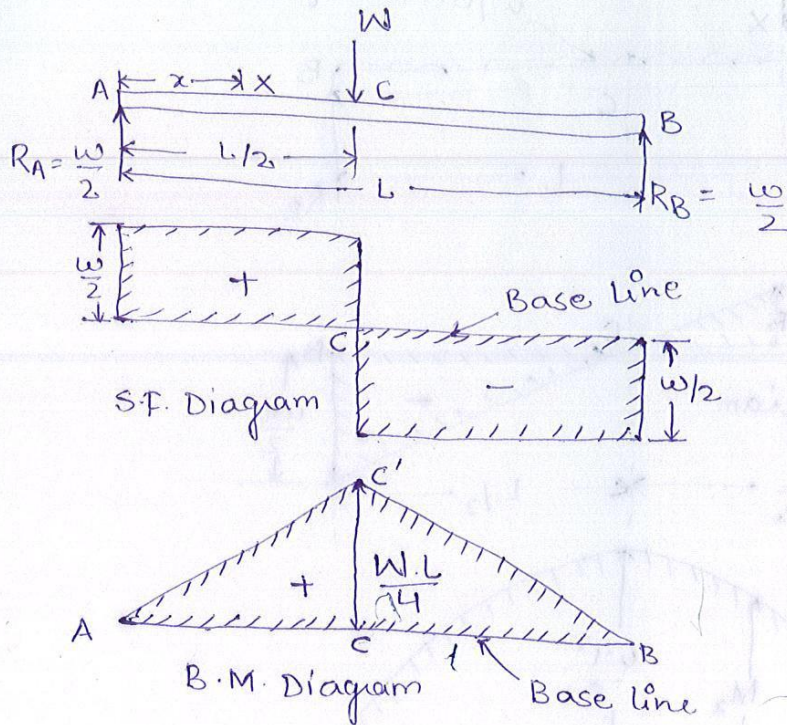
$$\text{S.F. at B} = 0$$

$$\text{S.F. at A} = \frac{1}{2} \times 6 \times 4 = 12$$

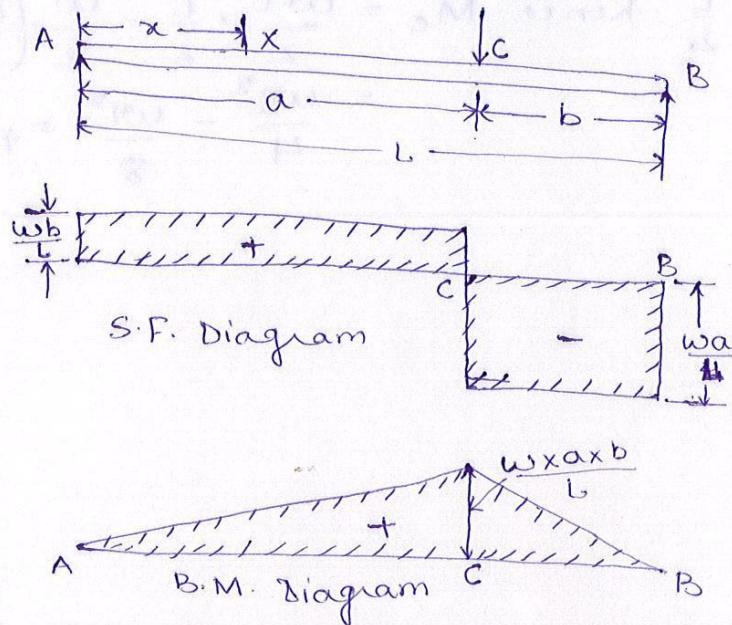
$$\text{B.M. at B} = 0$$

$$\begin{aligned} \therefore \text{M at A} &= \left(\frac{1}{2} \times 4 \times 6\right) \times 3 \\ &= 12 \times 2 = 24 \end{aligned}$$

Simply supported beam carrying a Pt load at centre of beam

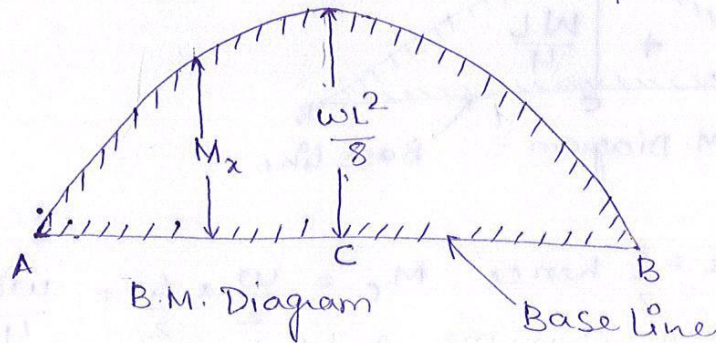
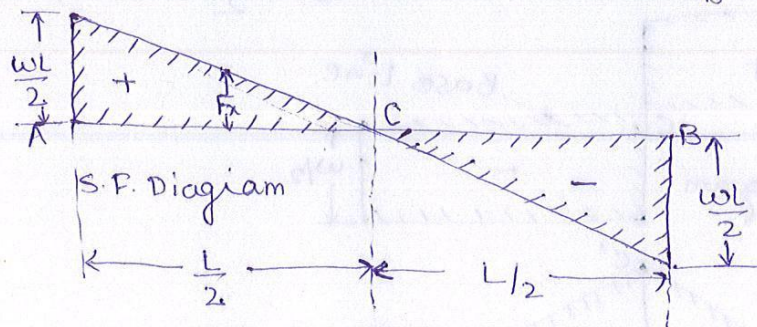
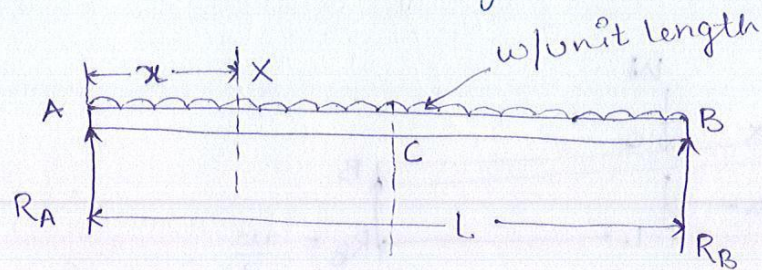


B.M. at C , $x = \frac{L}{2}$ hence $M_C = \frac{W}{2} \times \frac{L}{2} = \frac{WL}{4}$
 Simply supported beam carrying a pt load at other location than centre



Bending Moment at C , $x = a$ hence $M_C = \frac{W \cdot b \cdot a}{L}$
 $= \frac{W \cdot a \cdot b}{L}$

Simply supported beam carrying udl at whole span.

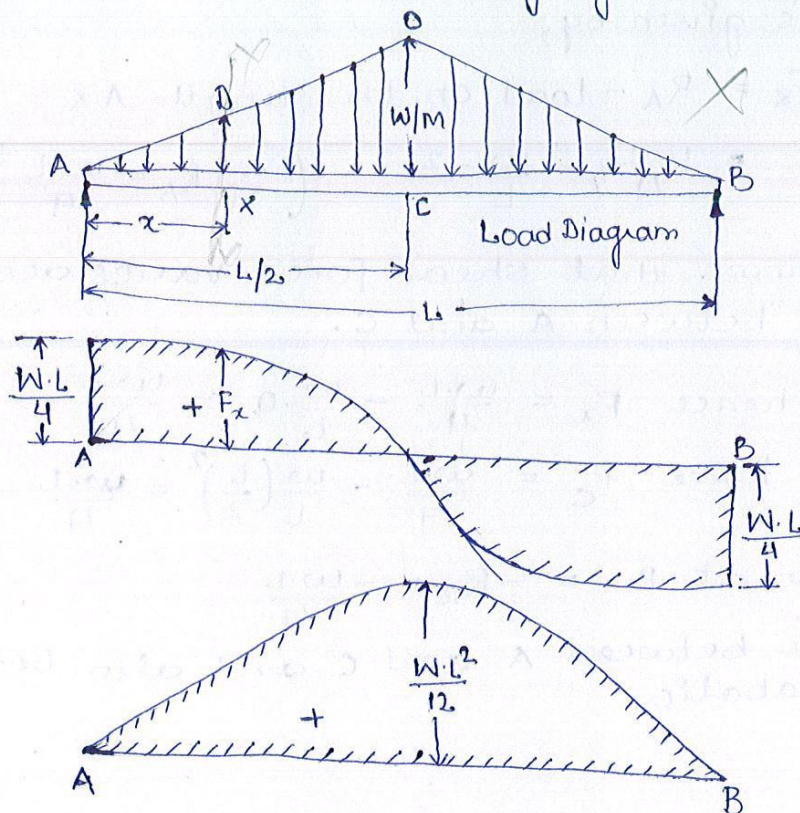


B.M. at C, $x = \frac{L}{2}$ hence $M_C = \frac{w \cdot L}{2} \cdot \frac{L}{2} - \frac{w}{2} \left(\frac{L}{2} \right)^2$

$$= \frac{wL^2}{4} - \frac{wL^2}{8} = +\frac{wL^2}{8}$$

Simply Supported beam carrying a UVL.

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The reaction at the supports will be equal and their magnitude will be half the total load on entire length, as the load is symmetrical on the beam.

But total load on the beam = Area of load diagram ABO

$$= \frac{AB \times CO}{2} = \frac{L \times w}{2} \text{ or } \frac{wL}{2}$$

$$\therefore R_A = R_B = \text{Half the total load}$$

$$= \frac{1}{2} \left(\frac{wL}{2} \right) = \frac{wL}{4}$$

Consider any section X between A and C at a distance x from end A.

The rate of loading at X

= Vertical distance XD in load diagram

$$= \left(\frac{x}{\frac{L}{2}} \right) \cdot w \quad \left[\because \frac{CO}{\left(\frac{L}{2}\right)} = \frac{XD}{x} \therefore XD = \frac{x \times CO}{\left(\frac{L}{2}\right)} = \frac{2x}{L} \right]$$

$$= \frac{2w}{L} \cdot x$$

Now load on the length AX of the beam = Area of load diagram AXD

$$= \frac{x \cdot XD}{2} = \frac{x \cdot \frac{2w}{L} \cdot x}{2} = \frac{wx^2}{L}$$

This load is acting at a distance of $\frac{x}{3}$ from X

Now S.F. at X is given by,

$$F_x = R_A - \text{load on the length } AX \\ = \frac{w \cdot L}{4} - \frac{w}{L} x^2 \quad \left(\because R_A = \frac{w \cdot L}{4} \right) \dots (i)$$

Equation (i) shows that shear force varies according to parabolic law between A and C.

$$\text{At A, } x=0 \text{ hence } F_x = \frac{w \cdot L}{4} - \frac{w}{L} \cdot 0 = \frac{w \cdot L}{4}$$

$$\text{At C, } x = \frac{L}{2} \text{ hence } F_c = \frac{w \cdot L}{4} - \frac{w}{L} \left(\frac{L}{2} \right)^2 = \frac{w \cdot L}{4} - \frac{w \cdot L}{4} = 0$$

$$\text{The shear force at B} = -R_B = -\frac{w \cdot L}{4}$$

The shear force between A and C and also between C and B is parabolic.

B.M. Diagram

The bending moment is zero at A and B

The B.M. at X is given by,

$$M_x = R_A \cdot x - \text{Load of length } AX \cdot \frac{x}{3} \\ = \frac{w \cdot L}{4} \cdot x - \frac{w}{L} x^2 \cdot \frac{x}{3} = \frac{w \cdot L}{4} \cdot x - \frac{w}{36} x^3 \dots (ii)$$

Equation (ii) shows that B.M. between A & C varies acc. to cubic law

$$\text{At A, } x=0 \text{ hence } M_x = 0$$

$$\text{At C, } x = \frac{L}{2} \text{ hence } M_c = \frac{w \cdot L}{4} \cdot \frac{L}{2} - \frac{w}{36} \left(\frac{L}{2} \right)^3 \\ = \frac{w \cdot L^2}{8} - \frac{w L^2}{24} = \frac{3w \cdot L^2 - w L^2}{24} = \frac{w L^2}{12}$$

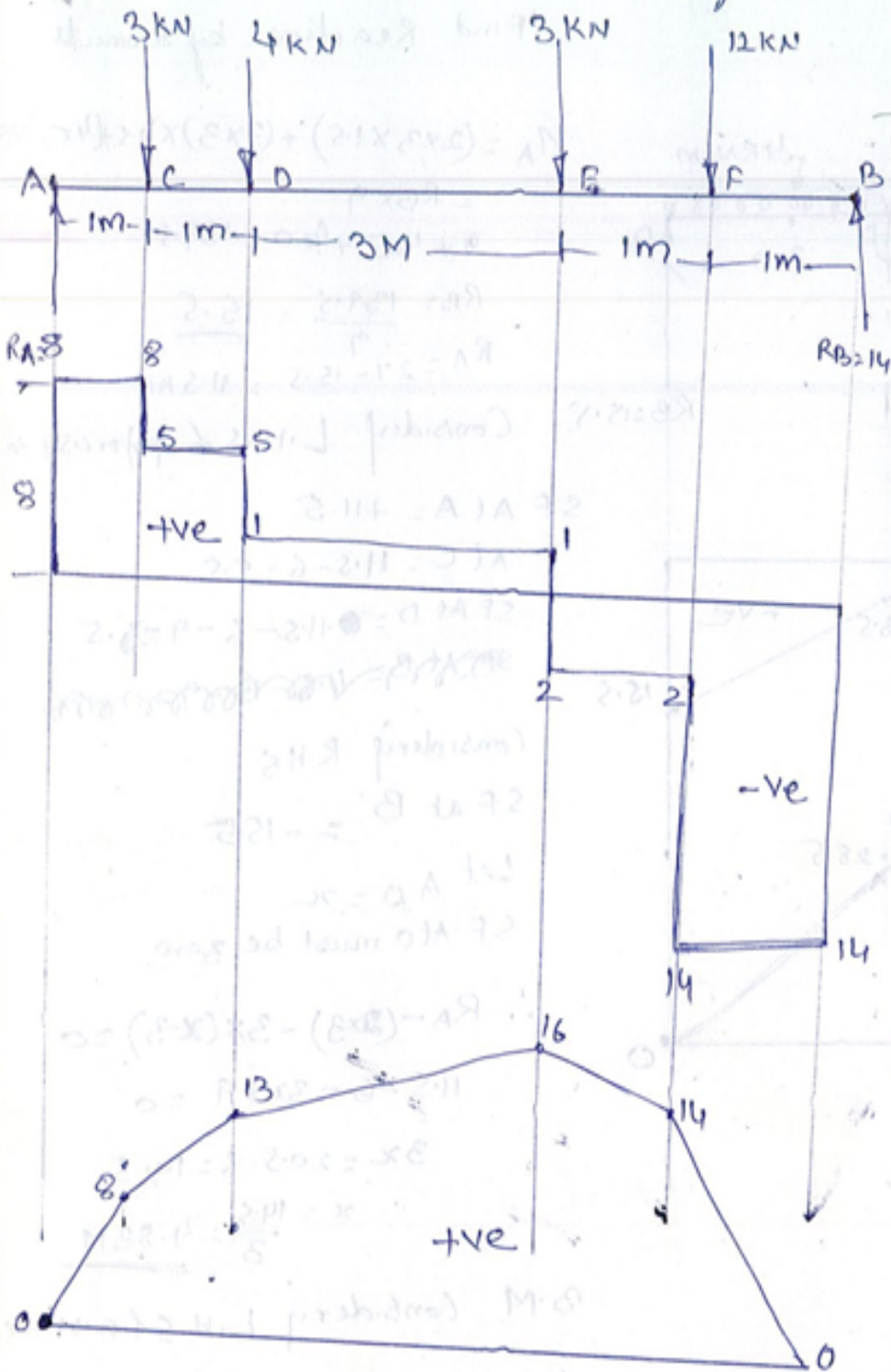
The maximum B.M. occurs at the centre of the beam, where S.F. becomes zero after changing its sign.

$$\therefore \text{Max B.M. is at C, } M_c = \frac{w L^2}{12}$$

$$\text{But total Load on the beam, } w = \frac{w L}{2}$$

$$\therefore \text{Maximum B.M.} = \frac{w L}{2} \cdot \frac{L}{6} = \frac{w \cdot L^2}{12}$$

② Draw S.F & B.M.D of a simply supported beam as shown in fig.



Find Reaction by taking Moments

$$M_A = 3 \times 1 + 4 \times 2 + 3 \times 5 + 12 \times 6 = 77 \text{ kNm}$$

$$\uparrow R_B = 3 + 8 + 15 + 72 = 98$$

$$R_B = \frac{98}{7} = 14$$

$$R_B = 14 \quad R_A = 22 - 14 = 8$$

Considering L.H.S & $\uparrow$ +ve
 $\downarrow$ -ve

$$\text{S.F at A} = 8$$

$$B = 8 - 3 = 5$$

$$D = 5 - 4 = 1$$

$$E = 1 - 3 = -2$$

$$F = -2 - 12 = -14$$

$$\text{At B} = -14$$

Considering L.H.S & Clock
wise B.M +ve

$$\text{B.M at A} = 0$$

$$\text{At C} = 8 \times 1 = 8$$

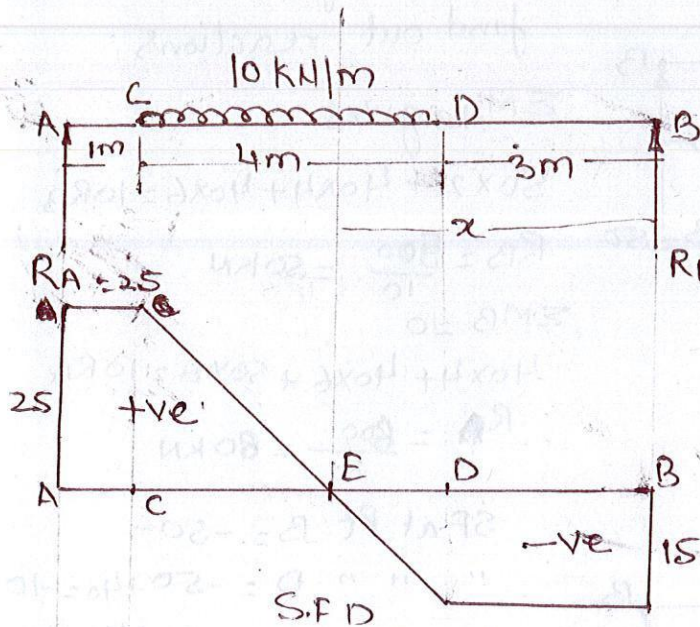
$$\text{At D} = 8 \times 2 - 3 \times 1 = 13$$

$$\text{At E} = 8 \times 5 - 3 \times 4 - 4 \times 3 = 40 - 12 - 12 = 16$$

$$\begin{aligned} \text{At F} &= 8 \times 6 - 3 \times 1 - 4 \times 4 \\ &\quad - 3 \times 5 \\ &= 48 - 3 - 16 - 15 = 14 \end{aligned}$$

$$\begin{aligned} \text{At B} &= 8 \times 7 - 12 \times 1 - 3 \times 2 - 4 \times 5 \\ &\quad - 3 \times 6 \\ &= 56 - 12 - 6 - 20 - 18 = 0 \end{aligned}$$

Draw the SFD and bending moment diagram for the beam as shown in fig. Also calculate the max B.M on the section.



Reactions R_A & R_B can be obtained as

$\Sigma M_A = 0$ gives

$$40 \times 3 = R_B \times 8$$

$$R_B = \frac{120}{8} = 15 \text{ kN}$$

$\Sigma M_B = 0$ gives

$$40 \times 5 = R_A \times 8$$

$$R_A = \frac{200}{8} = 25 \text{ kN}$$

$$\text{S.F at B} = -15$$

$$\text{SF at D} = -15$$

$$\text{SF at C} = -15 + 40 = 25$$

$$\text{SF at A} = 25$$

SFD is as shown in fig

Point E can be found out by writing the eqn of SF at E and equating it to zero

$$\text{SF at E} = -15 + 10(x-3) = 0$$

$$-15 + 10x - 30 = 0$$

$$10x = 45 \text{ or } x = 4.5$$

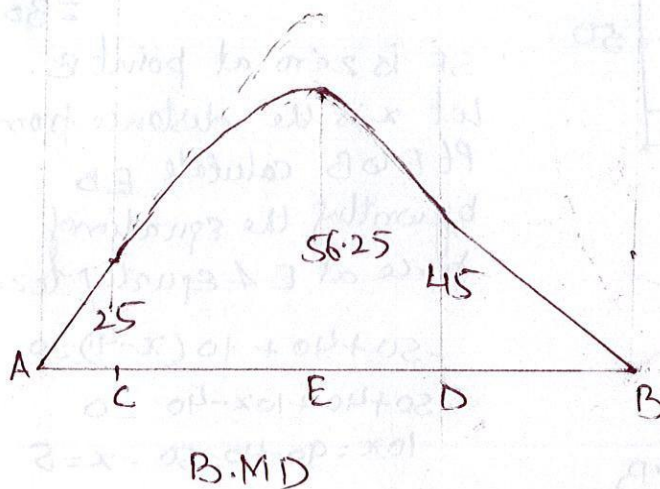
Now B.M at B = 0

$$\text{B.M at D} = 15 \times 3 = +45$$

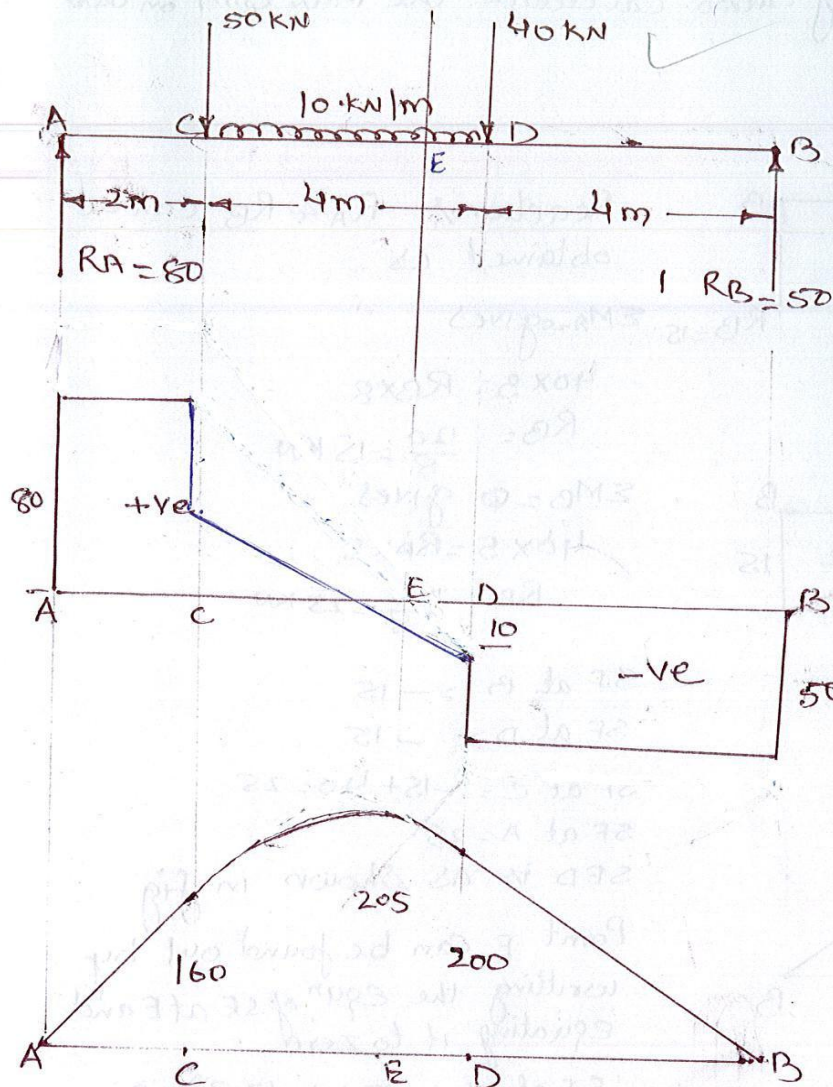
$$\text{B.M at E} = +15 \times 4.5 - 10 \times 1.5 \times 1.5 = 67.5 - 22.5 = 45$$

$$\text{B.M at A} = 0$$

B.M from B to D and from A to C governs at line load and from C to D will governs parabolic load.



Draw the SFD & BMD for the beam shown in fig



First of all we will find out reactions

$\sum M_A = 0$ gives

$$50 \times 2 + 40 \times 4 + 40 \times 6 = 10 R_B$$

$$R_B = \frac{500}{10} = 50 \text{ kN}$$

$\sum M_B = 0$

$$40 \times 4 + 40 \times 6 + 50 \times 8 = 10 R_A$$

$$R_A = \frac{600}{10} = 60 \text{ kN}$$

SF at Pt B = -50

" " " D = -50 + 40 = -10

" " " C = -50 + 40 + 40 = 30

SF is zero at point E

Let x is the distance from Pt E to B calculate EB

by writing the equation of force at E & equating to zero

$$-50 + 40 + 10(x - 4) = 0$$

$$-50 + 40 + 10x - 40 = 0$$

$$10x = 90 - 40 = 50 \quad x = 5$$

Now BM at pt B = 0

" " " " D = + ... $50 \times 4 = 20$

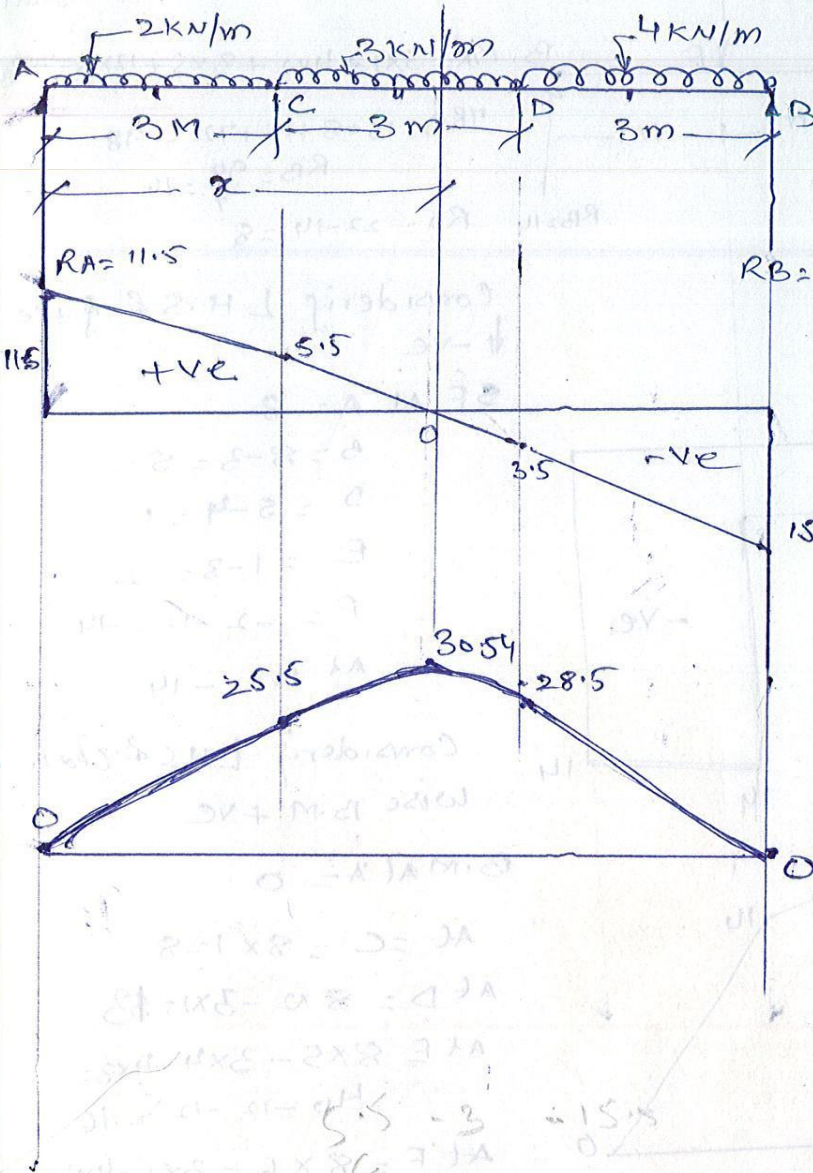
E = + ... $50 \times 5 - 10 \times 5 = 250 - 50 = 200$

C = $50 \times 8 - 40 \times 4 - 40 \times 2 = 160$

A = $50 \times 10 - 40 \times 6 - 40 \times 4 - 50 \times 2 = 0$

Draw the S.F & B.M.D. of a simply supported beam showing

Find Reaction by moment



$$M_A = (2 \times 3 \times 1.5) + (3 \times 3) \times 4.5 + (4 \times 3) \times 6$$

$$= R_B \times 9$$

$$9 + 40.5 + 72 = 121.5$$

$$R_B = \frac{121.5}{9} = 13.5$$

$$R_A = 27 - 13.5 = 13.5 \text{ kN}$$

Considering L.H.S & ↑ forces +ve

$$\text{SF at A} = +11.5$$

$$\text{At C} = 11.5 - 6 = 5.5$$

$$\text{SF at D} = 11.5 - 6 - 9 = -3.5$$

Considering R.H.S

$$\text{SF at B} = -13.5$$

Let $AO = x$

SF at O must be zero

$$\therefore R_A - (2 \times x) - 3 \times (x - 3) = 0$$

$$11.5 - 2x - 3x + 9 = 0$$

$$3x = 20.5 - 6 = 14.5$$

$$x = \frac{14.5}{3} = 4.83 \text{ m}$$

B.M Considering L.H.S & C.W +ve

$$\text{B.M at A} = 0$$

$$\text{At C} = 11.5 \times 3 - (2 \times 3) \times 1.5$$

$$34.5 - 9 = 25.5$$

$$\text{At D} = 11.5 \times 6 - (2 \times 3) \times 4.5$$

$$- (3 \times 3) \times 1.5$$

$$= 69 - 27 - 13.5 = 28.5 \text{ kNm}$$

$$\text{At B} = 11.5 \times 9 - (2 \times 3) \times 7.5 - (3 \times 3) \times 4.5$$

$$- 4 \times 3 \times 1.5$$

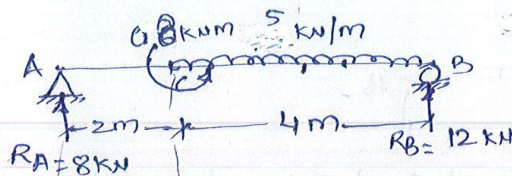
$$= 103.5 - 45 - 40.5 - 18 = 0$$

$$\text{At O} = 11.5 \times 4.83 - (2 \times 4.83) \times 2.415$$

$$- (3 \times 1.83) \times 1.83$$

$$55.54 - 19.98 - 5.023 = 30.54$$

Draw the SFD & BMD for the beam shown in fig



$$R_B \times 6 + 8 - 20 \times 4 = 0 \quad R_B = \frac{80 - 8}{6} = 12$$

$$R_A \times 6 - 8 - 20 \times 2 = 0$$

$$R_A = \frac{48}{6} = 8 \text{ kN}$$

$$\text{SF at B} = -12$$

$$\text{SF just R to C} = -12 + 20 = 8$$

$$\text{SF " L " C} = -12 + 20 = 8$$

$$\text{SF at a dist } x \text{ from B}$$

$$-12 + 5x = 0$$

$$x = \frac{12}{5} = 2.4 \text{ m}$$

$$\text{BM at B} = 0$$

$$\text{BM just R to C} = 12 \times 4 - 20 \times 2 = 8$$

$$\text{BM " L " C} = 12 \times 4 - 20 \times 2 + 8 = 16$$

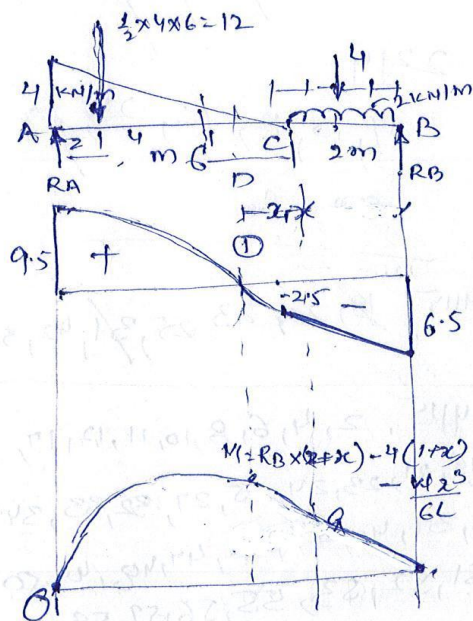
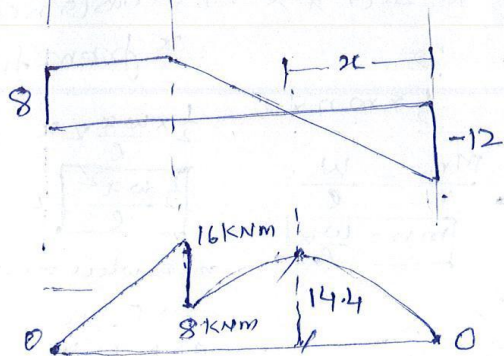
$$\text{BM at A} = 12 \times 6 + 8 - 20 \times 4$$

$$72 + 8 - 80 = 0$$

$$\text{BM at } 2.4 \text{ from B } \left(R_B x - \frac{w x^2}{2} \right)$$

$$= 12 \times 2.4 - (5 \times 2.4) \times 1.2$$

$$28.8 - 14.4 = 14.4$$



$$\sum M_A = 0 \quad 12 \times 2 + 4 \times 4 = 8 R_B$$

$$R_B = \frac{52}{8} = 6.5 \text{ kN}$$

$$R_A = 9.5$$

$$\text{SF at B} = -6.5 \quad \left(\uparrow + \downarrow R_B + w x \right)$$

$$\text{" just R to C} = -6.5 + 4 = -2.5$$

$$\text{" " " L " " } = -2.5$$

$$\text{SF at a dist } x \text{ from pt C}$$

$$= -R_B + 2x + \frac{4}{6} x x \times \frac{1}{2}$$

$$= -6.5 + 4 + \frac{4}{12} x^2 = -6.5 + 4 + \frac{1}{3} x^2$$

$$\text{At C when } x = 0 \quad \text{SF} = -6.5 + 4 = -2.5$$

$$\text{At A " } x = 6 \quad \text{"} = -6.5 + 4 + \frac{1}{3} \times 36 = 12$$

$$\text{SF at (1)} = 16 - 6.5 = 9.5$$

$$-R_B + \dots + \frac{4}{6} x x \times \frac{x}{2}$$

$$-6.5 + 4 + \frac{x^2}{3} = 0$$

$$\frac{x^2}{3} = 2.5 \quad x^2 = 3 \times 2.5$$

$$x = \sqrt{3 \times 2.5} = 2.7386 \text{ m}$$

$$\text{BM at B} = 0 \quad \text{at C} = 6.5 \times 2 - 4 \times 1 = 9$$

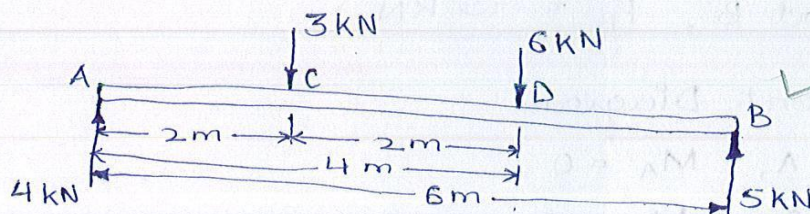
$$\text{At D} = 6.5 \times 4 - 4 \times 3.7386 - \frac{1}{2} \times 1.8257 \times 2.7386 = 2.5$$

$$= 30.8 - 14.95 - 2 = 13.85$$

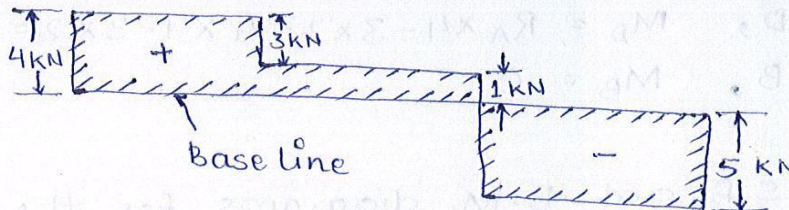
$$30.8 - 14.95 - 2.28 = 13.57$$

$$\times \frac{2.7386}{3}$$

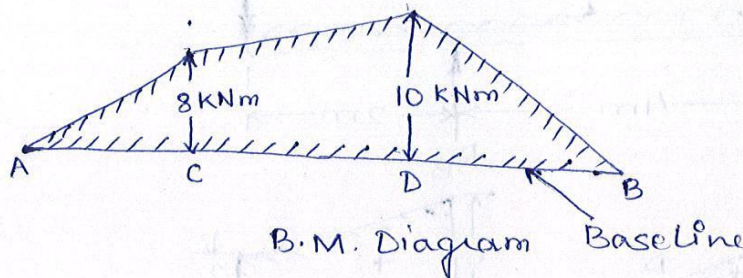
1. Draw the shear force and bending moment diagrams for the given beam.



Solⁿ:



S.F. Diagram



B.M. Diagram

First calculate the reactions R_A and R_B .
Taking moments of the force About A, we get

$$R_B \times 6 = 3 \times 2 + 6 \times 4 = 30$$

$$\Rightarrow R_B = 5 \text{ kN}$$

$$\therefore R_A = \text{Total Load on beam} - R_B \\ = (3+6) - 5 = 4 \text{ kN.}$$

Shear Force Diagram

Shear force at A, $F_A = +R_A = +4 \text{ kN}$

Shear force between A and C is constant and equal to +4 kN.

Shear force at C, $F_C = +4 - 3.0 = +1 \text{ kN}$

Shear force between C and D is constant and equal to +1 kN.

Shear force at D, $F_D = +1 - 6 = -5 \text{ kN}$

The shear force between D and B is constant and equal to -5 kN

Shear force at B, $F_B = -5 \text{ kN}$

Bending Moment Diagram

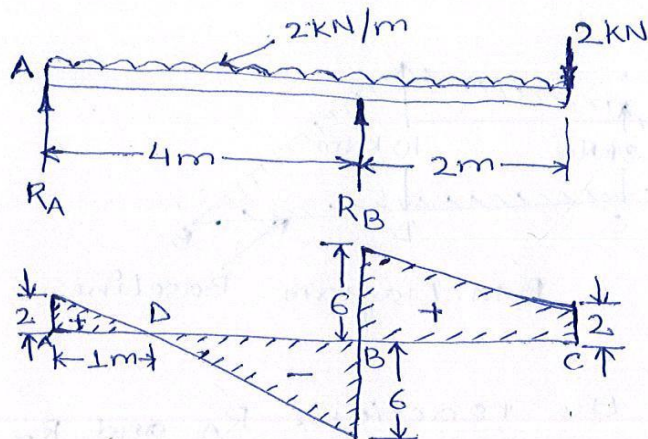
B.M. at A, $M_A = 0$

B.M. at C, $M_C = R_A \times 2 = 4 \times 2 = +8 \text{ kNm}$

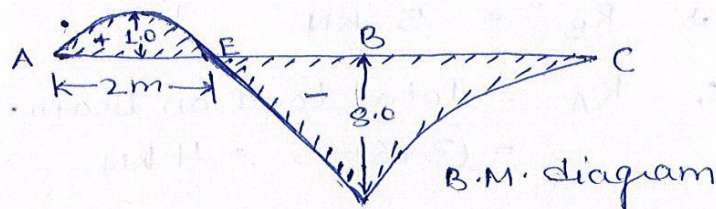
B.M. at D, $M_D = R_A \times 4 - 3 \times 2 = 4 \times 4 - 3 \times 2 = +10 \text{ kNm}$

B.M. at B, $M_B = 0$

2. Draw the S.F. and B.M. diagrams for the given overhanging beam.



S.F. Diagram



B.M. diagram

First calculate the reactions R_A & R_B .

Taking moments of all forces about A, we get

$$R_B \times 4 = 2 \times 6 \times 3 + 2 \times 6 = 36 + 12 = 48$$

$$R_B = 12 \text{ kN}$$

$$\therefore R_A = \text{Total Load} - R_B = (2 \times 6 + 2) - 12 = 2 \text{ kN}$$

S.F. Diagram

$$\text{S.F. at A} = +R_A = +2 \text{ kN}$$

- i) The S.F. at any section between A and B at a distance x from A is given by,

$$F_x = +R_A - 2 \times x = 2 - 2x \quad \text{--- (i)}$$

$$\text{At A, } x=0 \text{ hence } F_A = 2 - 2 \times 0 = 2 \text{ kN}$$

$$\text{At B, } x=4 \text{ hence } F_B = 2 - 2 \times 4 = -6 \text{ kN}$$

The S.F. between A and B varies according to straight line law. At A, S.F. is positive and at B, S.F. is negative. Hence between A and B, S.F. is zero. The point of zero S.F. is obtained by substituting $F_x = 0$ in eq. (i)

$$0 = 2 - 2x \text{ or } x = \frac{2}{2} = 1 \text{ m.}$$

The S.F. is zero at point D. Hence distance of D from A is 1 m.

- ii) The S.F. at any section between B and C at a distance x from A is given by,

$$\begin{aligned} F_x &= +R_A - 2 \times 4 + R_B - 2(x-4) \\ &= 2 - 8 + 12 - 2(x-4) = 6 - 2(x-4) \end{aligned}$$

$$\text{At B, } x=4 \text{ hence } F_B = 6 - 2(4-4) = +6 \text{ kN}$$

$$\text{At C, } x=6 \text{ hence } F_C = 6 - 2(6-4) = 2 \text{ kN}$$

B.M. Diagram

B.M. at A is zero

- i) B.M. at any section between A and B at a distance x from A is given by,

$$M_x = R_A \times x - 2 \times x \times \frac{x}{2} = 2x - x^2$$

The above equation shows that B.M. between A and B varies according to parabolic law.

$$\text{At A, } x=0 \text{ hence } M_A = 0$$

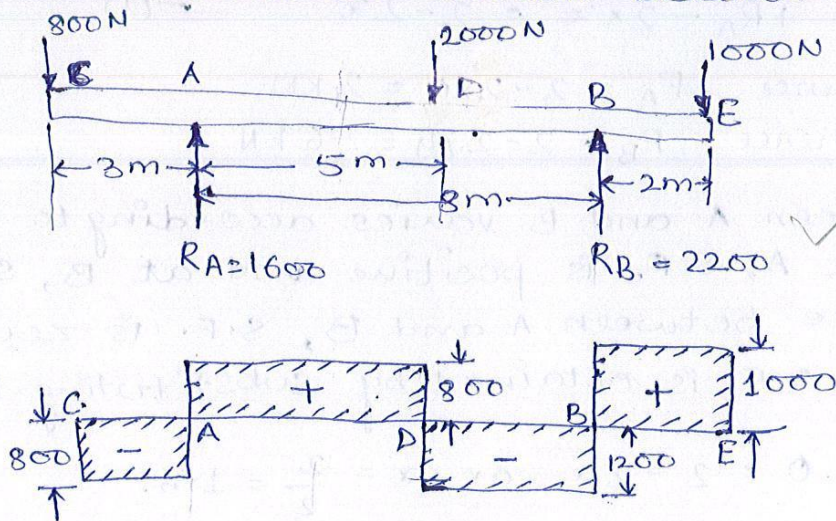
$$\text{At B, } x=4 \text{ hence } M_B = 2 \times 4 - 4^2 = -8 \text{ kNm}$$

Max. B.M. is at D where S.F. is zero after changing sign.

$$\text{At D, } x=1 \text{ hence } M_D = 2 \times 1 - 1^2 = 1 \text{ kNm}$$

The B.M. at C is zero. The B.M. also varies between B and C according to parabolic law.

3. Draw the S.F. and B.M. diagrams for the beam which is loaded as shown below.



S.F. Diagram

S.F. at C = -800 N

S.F. between C and A remains -800 N

S.F. At A = $-800 + R_A = -800 + 1600 = 800\text{ N}$

S.F. between A and D remains $+800\text{ N}$

S.F. at D = $+800 - 2000 = -1200\text{ N}$

S.F. between D and B remains -1200 N

S.F. at B = $-1200 + R_B = -1200 + 2200 = +1000\text{ N}$

S.F. between B and E remains $+1000\text{ N}$

B.M. Diagram

B.M. at C = 0

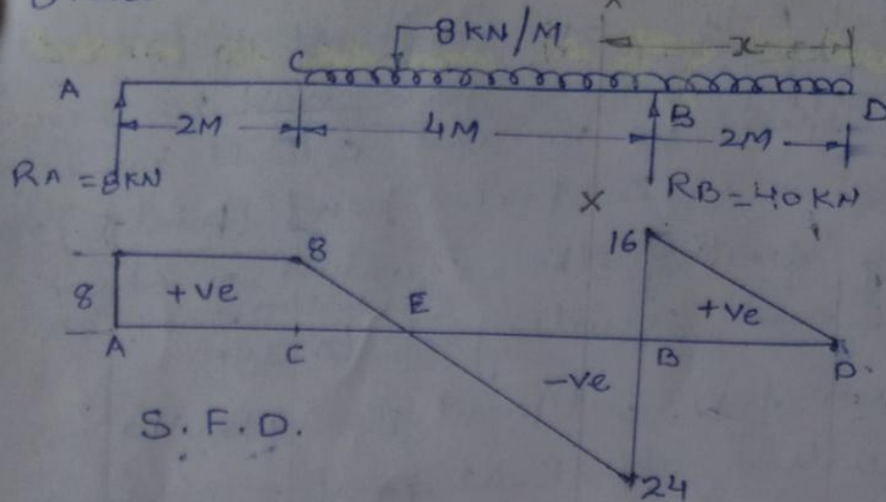
B.M. at A = $-800 \times 3 = -2400\text{ Nm}$

B.M. at D = $-800 \times (3+5) + R_A \times 5 = -6400 + 8000 = +1600\text{ Nm}$

B.M. at B = $-1000 \times 2 = -2000\text{ Nm}$

B.M. at E = 0

Draw the SFD and BMD of the overhanging beam as shown in fig.



Solution:

Considering the downward acting forces and anticlockwise moment to the right of the section positive

Firstly find out the reactions R_A & R_B

Taking moment of all the forces about point A & Equating it to zero.

$$\sum M_A = 0 \text{ gives } -48 \times 5 - 6 \times R_B = 0 \Rightarrow R_B = \frac{48 \times 5}{6} = 40 \text{ kN}$$

$$\text{From } \sum F_y = 0 \quad R_A + R_B = 8 \times 6 = 48 \quad R_A + 40 = 48 \Rightarrow R_A = 48 - 40 = 8 \text{ kN}$$

Now considering a section X-X at a distance x from D between B & D

$$\text{Equation of shear force at section X-X is}$$

$$\text{S.F at X-X} = wx = 8x$$

$$\text{SF at D where } x = 0 \quad \therefore \text{SF at D} = 0$$

$$\text{SF just right to pt B where } x = 2 \quad \text{SF is } wx = 8 \times 2 = 16 \text{ kN}$$

$$\text{SF at point B} = wx - 40 = 8 \times 2 - 40 = 8 \times 2 - 40 = -24 \text{ kN}$$

$$\text{Taking section X-X at a distance } x \text{ from D between point B and C is given by the equation } wx - 40$$

$$\text{SF at pt C where } x = 6 \quad 8 \times 6 - 40 = 8 \text{ kN}$$

There is no load between point C and A Hence shear force at point A = 8 kN

The equation of shear force between point D and B and between point B and C are the equations of straight line. Hence shear force is

shear force at point B is -24 and shear force at point C is 8 . Therefore shear force must be zero at any point between B and C will be zero. The location of zero shear force point (point E) may be found out by writing the equation of shear force at point E and equating it to zero.

Equation of shear force at point E $Wx - 40$
Equating the equation of SF at point E to zero we get

$$Wx - 40 = 0 \Rightarrow 8x = 40 \text{ and } x = \frac{40}{8} = 5 \text{ m.}$$

For Bending Moment Diagram: Taking a section x-x at a distance x from D between D and B. Equation of bending Moment at section x-x

$$-W \times x \times \frac{x}{2} = -wx^2/2$$

At Point D x is zero Hence B.M at point D = 0

At point B $x = 2$ Hence Bending moment at C

Now taking section x-x at a distance x from D between points B and C

$$\text{B.M at } x-x = -8x \times \frac{x}{2} + 40(x-2)$$

At pt B. $x = 2$ Hence B.M = $-8 \times 2 \times \frac{2}{2} + 40(2-2) = -16 \text{ kNm}$

At pt E $x = 5$ Hence B.M at E = $-8 \times 5 \times \frac{5}{2} + 40(5-2) = 20 \text{ kNm}$

At pt C $x = 6$ " " at C = $-8 \times 6 \times \frac{6}{2} + 40(6-2) = 16 \text{ kNm}$

B.M at a section x-x at a distance x from point D in between point A and C will be given by the equation

$$-W \times 6(x-3) + 40(x-2)$$

B.M at A where x is 8 m.

$$= -8 \times 6(8-3) + 40(8-2) = 240 - 240 = 0$$

The equations of B.M between points B & D and C & B are parabolic. Hence the B.M.D between points C & D will governs the parabolic law.

The equation of B.M between pts A and C is the equation of a straight line hence the B.M.D between points A and C governs the straight line law.

Shear force diagram and bending moment diagram are as shown in fig.

in the bending moment diagram point F is point of contraflexure.

Point of contraflexure: Point of contraflexure is the point in bending moment diagram where bending moment changes sign from +ve to -ve or viceversa.

The location of pt of contraflexure can be found out by writing the equation of bending moment at this point and equating the equation to zero.

The location of pt F can be determined by the equation

$$-8 \times x \times \frac{x}{2} + 40(x-2) = 0$$

$$-\frac{8x^2}{2} + 40x - 80 = 0$$

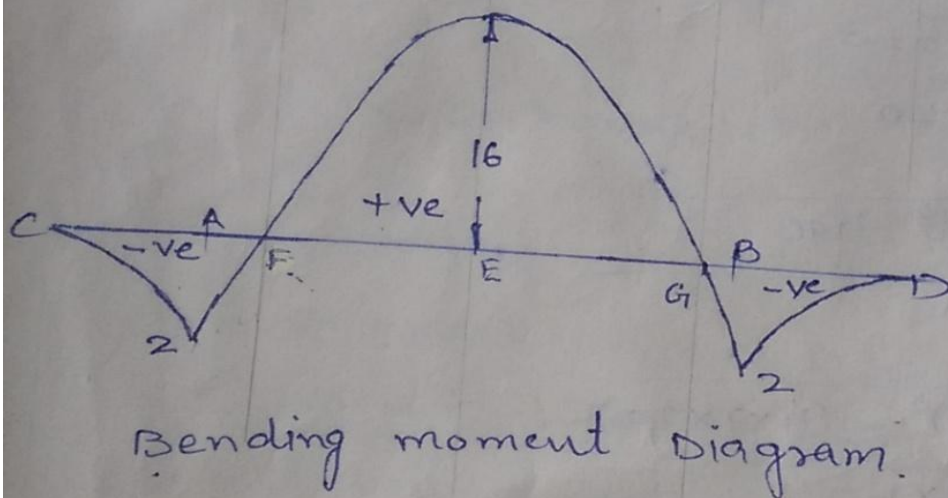
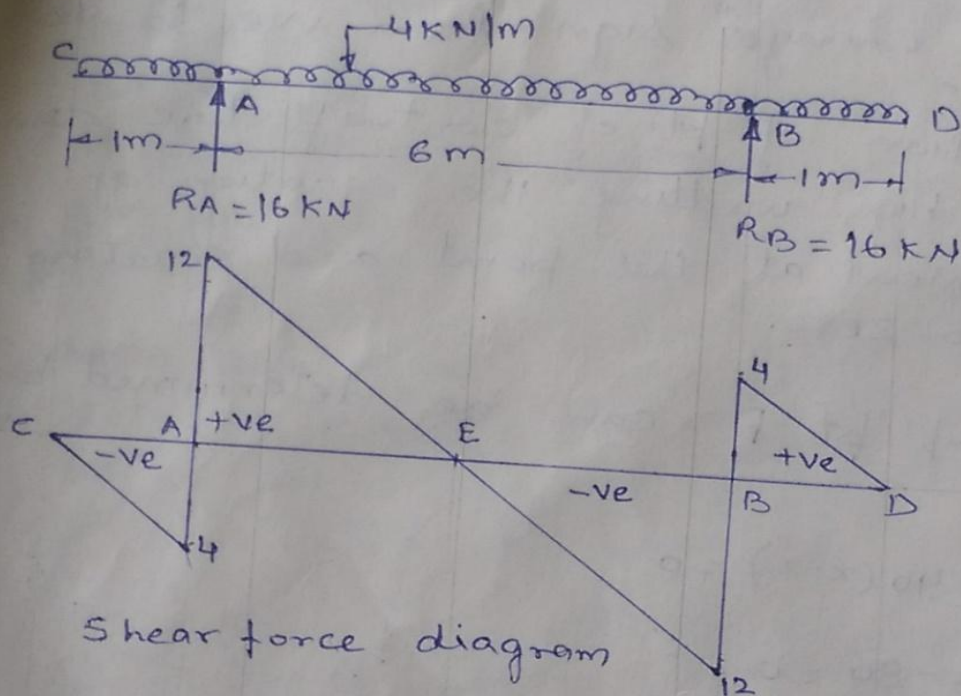
$$-4x^2 + 40x - 80 = 0$$

$$x^2 - 10x + 20 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$= \frac{10 \pm \sqrt{(-10)^2 - (4 \times 1 \times 20)}}{2 \times 1}$$

Draw the SFD and B.MD for the beam shown below also determine the location of point of contraflexures if any.



For finding out reactions R_A and R_B
 $\sum M_A = 0$ gives

$$4 \times 1 \times \frac{1}{2} + 6R_B - 4 \times 7 \times 3.5 = 0$$

$$= 2 + 6R_B - 98 = 0 \quad \text{or} \quad 6R_B = 98 - 2 = 96 \quad R_B = \frac{96}{6} = 16 \text{ kN}$$

From $\sum F_y = 0$

$$R_A = \text{Total Load} - R_B$$

$$R_A = 4 \times 8 - 16 = 16 \text{ kN}$$

Now considering the downward acting forces and anticlockwise moment to the right of the section are +ve

For Shear forces

Taking a section x-x at a distance x from D between B and D

$$\text{S.F at section x-x} = Wx$$

$$\text{At pt D } x=0 \text{ hence SF at D} = 0$$

S.F just right to the point B where $x=1\text{m}$

$$\text{SF " " " " " " " } = 4 \times 1 = 4 \text{ kN/m}$$

At a section x-x from a distance x from pt D between B and A

$$\text{S.F at x-x} = Wx - 16$$

$$\text{At Pt B } x=1 \text{ SF at Point where } x=1$$

$$= 4 \times 1 - 16 = -12$$

$$\text{SF just Right to pt A where } x=7$$

$$\therefore \text{SF " " " " A } = 4 \times 7 - 16 = 12 \text{ kN}$$

SF at a section x-x at a distance x from Point D in between pt A and is is given by the equation $Wx - 16 - 16$

$$\text{SF at A where } x=7$$

$$= 4 \times 7 - 32 = -4 \text{ kN}$$

$$\text{SF at point C where } x=8$$

$$= 4 \times 8 - 32 = 0$$

The equation of Shear forces at between all the points is equation of st line therefore the shear force diagram between all the pts will govern the st line law.

The Shear force between point A and B at a pt E will be zero. The location of pt E can be determined by writing the equation of SF at pt E and equating to zero. Let x be the distance of E from pt D.

$$\text{Hence DE} = x = Wxx - 16 = 0$$

$$4 \times x - 16 = 0 \text{ or } 4x = 16 \text{ or } x = \frac{16}{4} = 4\text{m}$$

For Bending moment

Take any Section x-x at a distance x from pt D between B & D

$$B.M \text{ at } x-x = -wx \times \frac{x}{2} = -\frac{wx^2}{2}$$

At pt D $x=0$ Hence B.M. $= -\frac{4 \times 0^2}{2} = 0$

at pt B $x=1$ B.M at B $= -\frac{4 \times 1^2}{2} = -2 \text{ kNm}$

B.M at a section x-x from D between A and B
 $= -wx \times \frac{x}{2} + 16 \times (x-1)$

B.M at E where $x=4$ is given by $-\frac{4 \times 4^2}{2} + 16(4-1)$
 $= -\frac{4 \times 16}{2} + 16 \times 3 = -32 + 48 = 16 \text{ kNm}$

B.M at any section x-x at a distance x from D between A and C is given by $-wx \times \frac{x}{2} + 16(x-1) + 16(x-7)$
B.M at A where $x=8$
 $= -\frac{wx^2}{2} + 16(x-1) + 16(x-7)$

B.M at A $= -\frac{4 \times 8^2}{2} + 16(8-1) + 16(8-7)$
 $= -128 + 112 + 16 = 0$

The equation of B.M between all the pts is parabolic hence bending moment diagram governs the parabolic law between all pts as shown in B.M.D

Point of contraflexure F and G will be at the same distances from pts C and D respectively because the loading on beam and the distance are similar

Hence the location of pt G i.e the distance DG will be given by

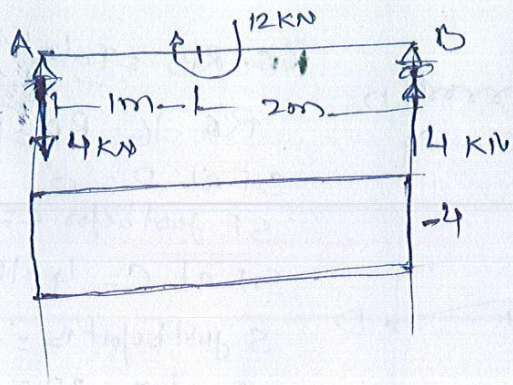
$$-wx \times \frac{x}{2} + 16(x-1) = 0 \quad \text{Hence } -\frac{4 \times x^2}{2} + 16(x-1) = 0$$

$$= -2x^2 + 16x - 16 = 0$$
$$= x^2 - 8x + 8 = 0$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{8 \pm \sqrt{(-8)^2 - (4 \times 1 \times 8)}}{2 \times 1}$$

$x =$



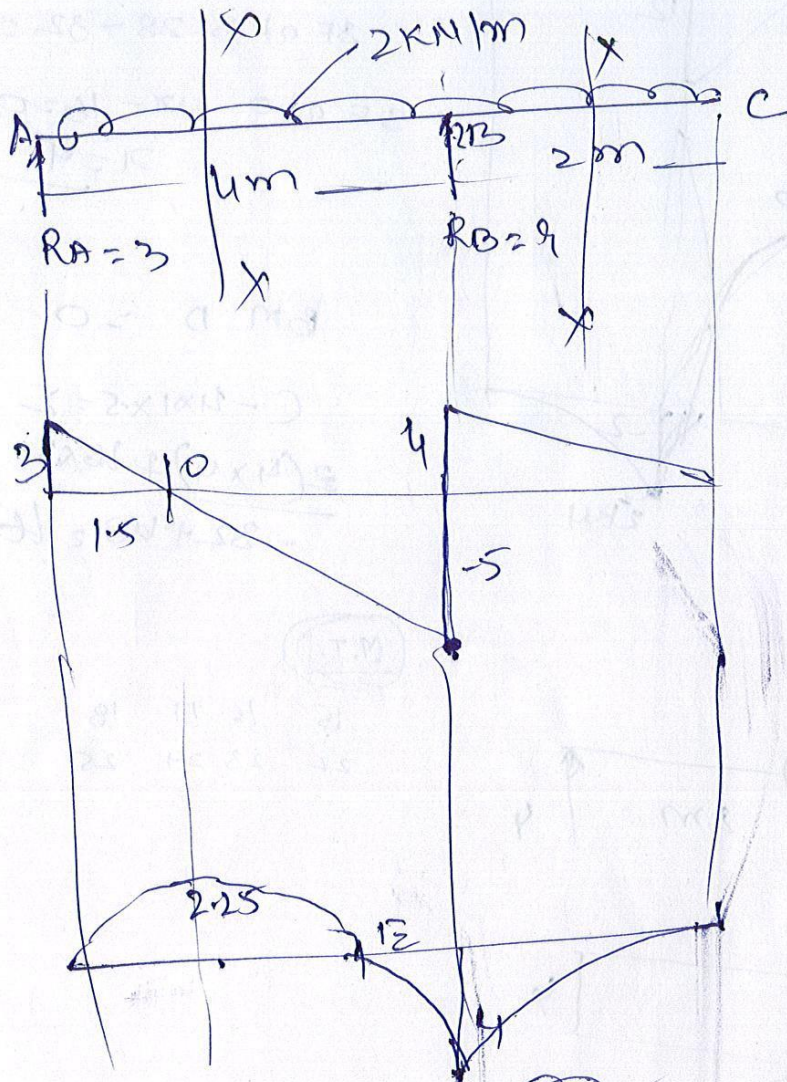
$$\sum M_A = 0$$

$$12 \text{ kN} - 3R_B = 0 \quad R_B = 4$$

$$\sum M_B = 0 \quad 12 \text{ kN} + 3R_A = 0$$

$$3R_A = -12$$

$$R_A = -4$$



SF at x-x from A

$$R_A - 2x$$

$$\text{at } A \quad x=0 \quad SF = 3 - 2 \times 0 = 3$$

just above B

$$SF = 3 - 2 \times 4 = -5$$

SFB SF at a sect

xx from A bet B & C

$$3 + 9 - 2 \times x$$

$$\text{at } x = 4$$

$$\frac{3 + 9 - 2 \times 4}{12 - 8} = 4$$

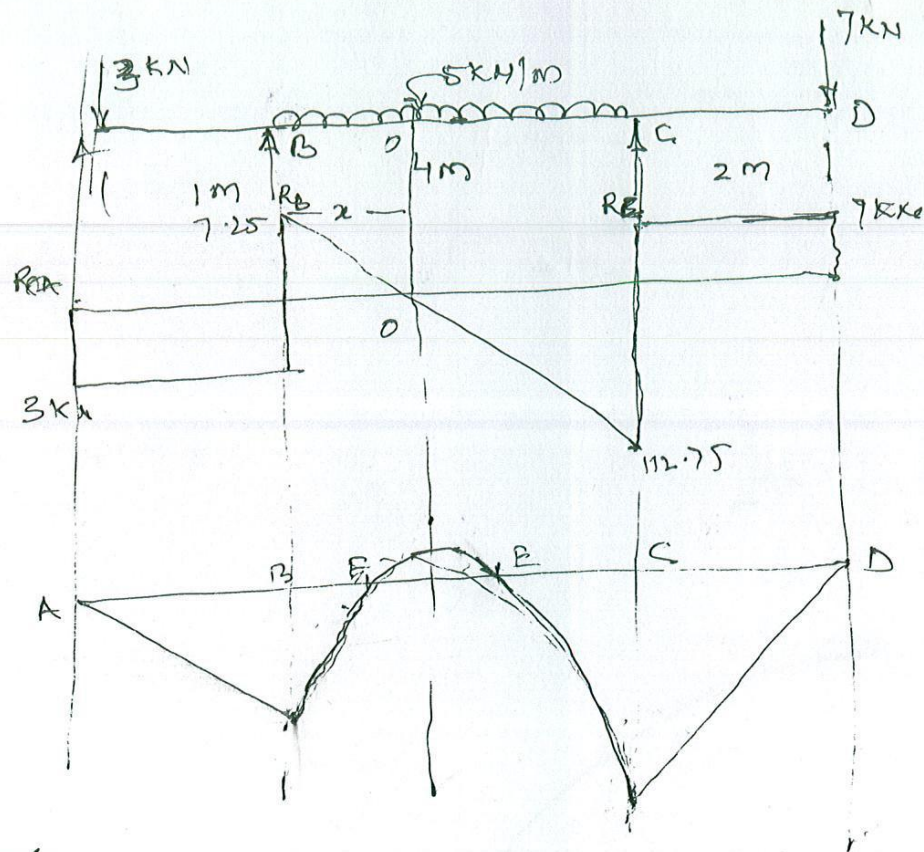
$$3x - x^2 = 0$$

$$25 \mid 415 \quad (2)$$

1, 4, 5, 6, 7, 9, 15, 17, 18, 19, 23

25, 32, 33, 34, 37, 38, 42, 43, 44

45, 51, 52, 53, 55, 56, 62



Reactions

$$M_C = R_B \times 4 = 3 \times 5 + 5 \times 4 \times 2$$

$$+ 7 \times 2$$

$$4R_B + 4 = 15 + 40$$

$$4R_B = 55 - 4 = 51$$

$$R_B = \frac{51}{4} = 12.75$$

$$R_A = 3 + 7 + 20 - 12.75 = 17.25$$

$$\text{SF at A} = -3 \text{ kN}$$

approach

$$\text{SF at B}_2 = -3 \text{ kN}$$

$$\text{SF at B}_2 = -3 + 12.75 = 9.75 \text{ kN}$$

Considering

$$\text{SF at D} = 7 \text{ kN}$$

$$\text{SF approach C} = 7 \text{ kN}$$

$$\text{SF at C} = 7 - R_C = 7 - 17.25 = -10.25 \text{ kN}$$

$$\text{SF at } 0 = 0 = -3 + R_B - 5x = 0$$

$$-3 + 12.75 - 5x = 0$$

$$x = \frac{9.75}{5} = 1.95 \text{ m}$$

$$\text{BM at A} = 0$$

$$B = -$$